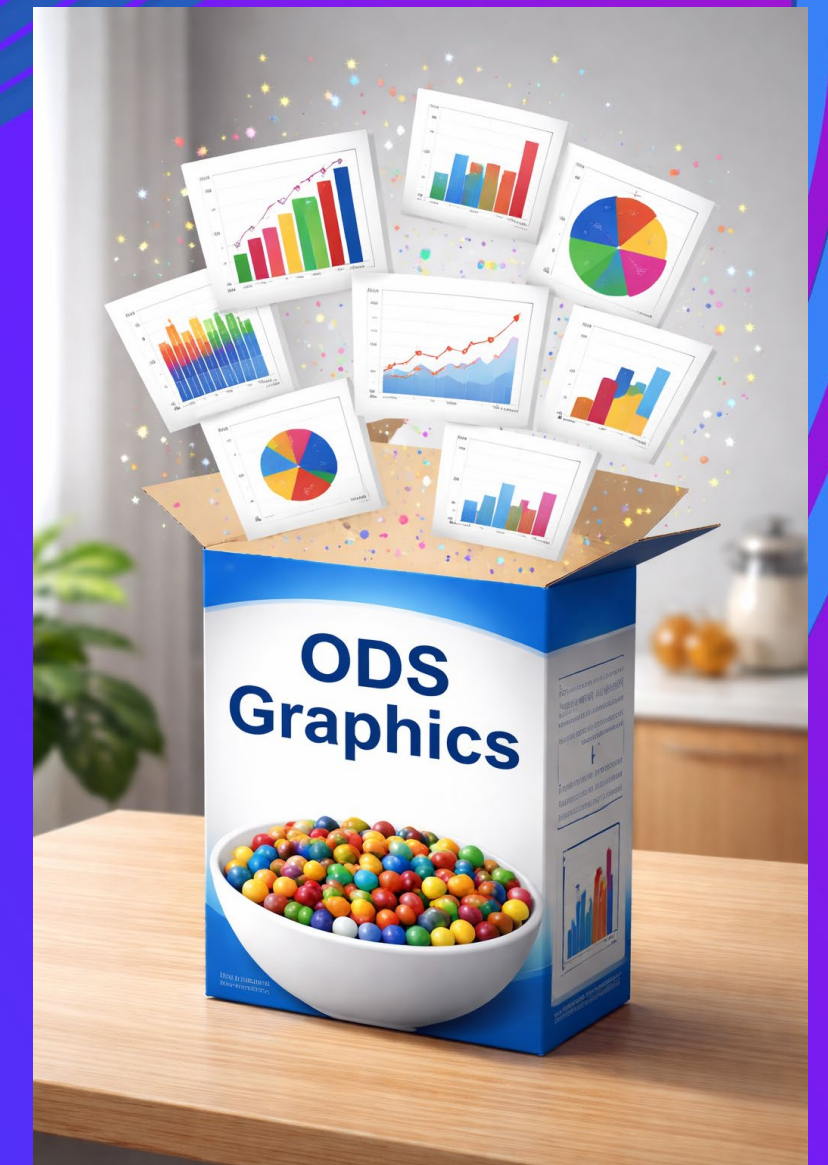


OOTB (Out of the Box) Data Visualizations with SAS:

An Introduction to what the ODS Graphics Statement Can Do for You



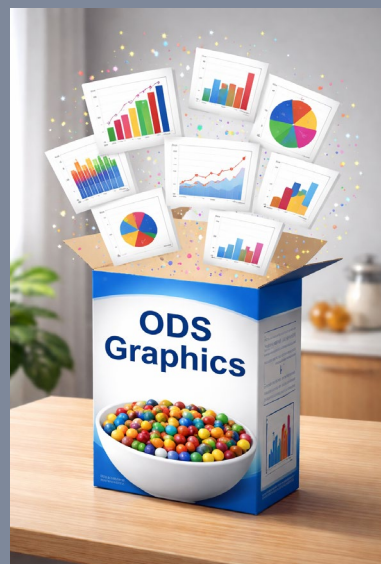
Louise S. Hadden, Western Users of SAS Software (WUSS)

Biography

Louise S. Hadden

Louise Hadden presented at her first SAS conference in 1996 and has never looked back, presenting at multiple conferences across the continent over the years. She supports file building and analytic programming for life sciences organizations, most frequently as a contractor for government agencies such as CMS and CDC, and specializes in reporting and data visualization.





Introduction



Introduction

In the beginning...

- Creating graphic outputs with SAS[®] software has been the subject of many a SAS[®] paper over the years
- SAS users have many options when it comes to statistical graphs: SAS/Graph, JMP, various modules such as IML, and the more recent SG (statistical graph) procedures.
- When the Output Delivery System (ODS) first arrived on the scene in SAS Version 7, it drew on a number of earlier developments SAS made to enhance output in general, including the use of templates, styles, and options for list and graphic outputs, created by both users and developers.



Introduction

The Dawn of ODS Graphics!

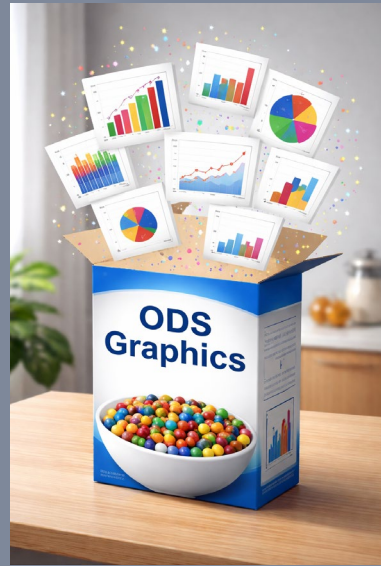
- SAS introduced a pre-production version of ODS GRAPHICS, a new graphics system which was template based.
- This new system was originally part of SAS/GRAPH, which is a separate SAS product
- In ODS GRAPHICS, many statistical procedures could produce one or more graphic, depending on options within the procedure, etc., simply by turning on the ODS GRAPHICS option.
- In SAS 9.2, ODS GRAPHICS moved toward production, with the introduction of the SG family of procedures.
- As with ODS, SAS developed templates for each statistical graphic procedure and option



Introduction

Tell Me More...

- There are many ODS statements in BASE SAS, including ODS [destination/tagset of your choice], ODS OUTPUT, ODS TRACE, etc.
https://documentation.sas.com/api/docsets/odsug/v_004/content/odsug.pdf?locale=en
- This presentation will hone in on the ODS Graphics statement, and its many wonderful options, and why you might want to add some OOTB ODS graphics to your toolbox.
- For example, with the context of a large scale data base relating to patient safety with thousands of data elements within different health systems, how can we:
 - Explore data elements graphically as single elements (univariate statistics and graphs)
 - Explore data elements graphically as elements in conjunction with other elements (multivariate statistics and graphs)
 - All before writing custom SG procedure code or learning Graphics Template Language?



The Case Study

Millions of Adverse Event records that need to be analyzed, and presented in Dashboards and Stakeholders



The Case Study

Patient Safety Organization Data

- Millions of adverse event records over 18 years
- Data collected using Common Formats developed by HHS
- Contains demographics and highly specific event data
- Thousands of variables
- Data is published in multiple dashboards by Event Type on an annual basis
- Need a quick way to determine what is relevant and safe for publication
- <https://psa.ahrq.gov/common-formats/npsd-dashboards>
- <https://psa.ahrq.gov/common-formats/about>

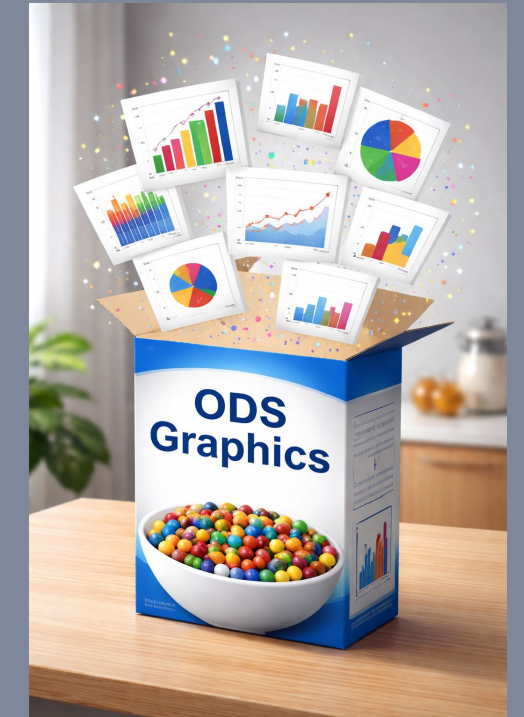


The Ersatz Data

SASHELP.HEART

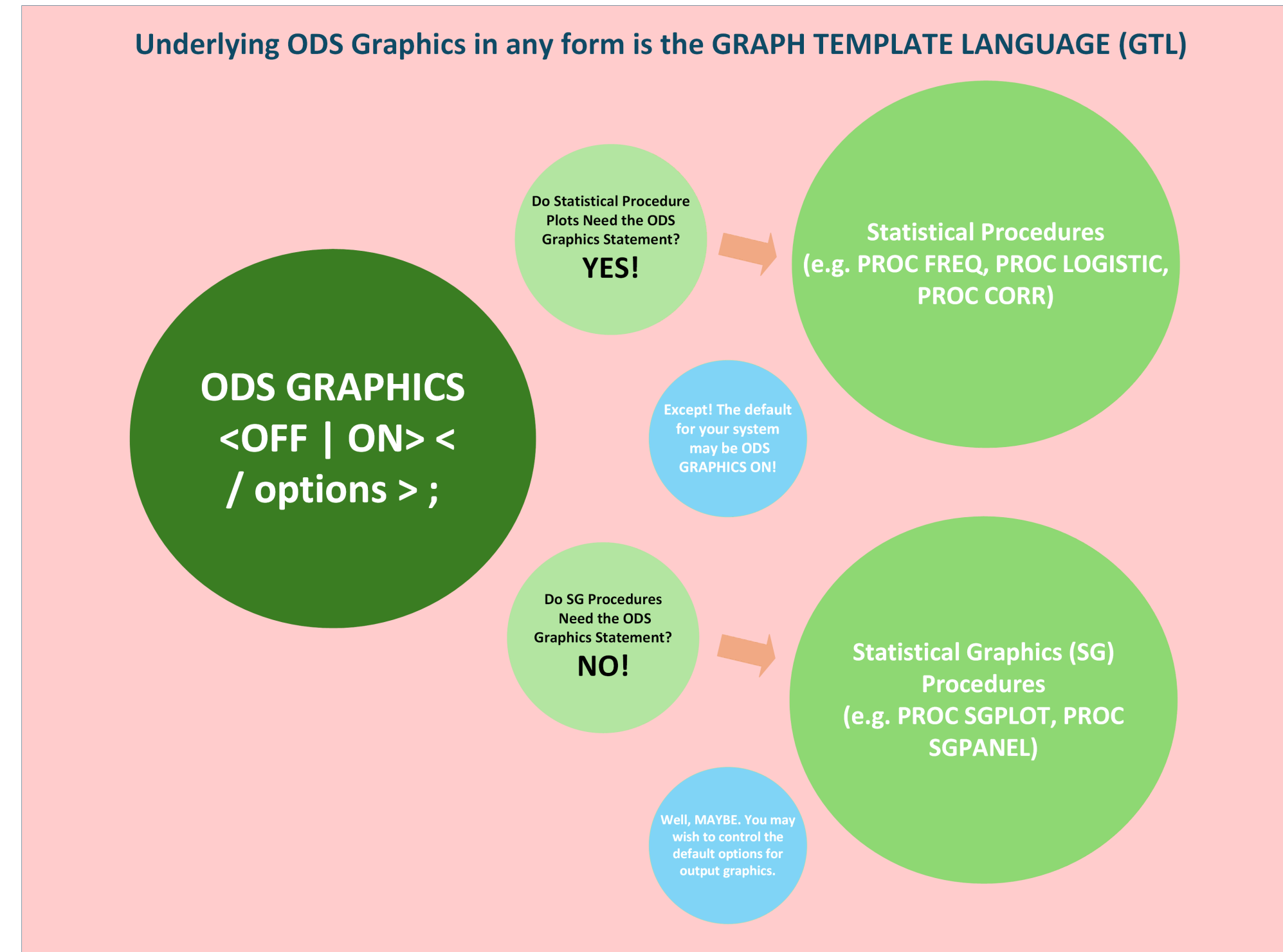
- The SASHELP.HEART data set is based on the Framingham Heart Study, started in 1948 in Framingham, MA.
- Data set contains demographics, medical data, and common risk factors for cardiovascular disease (CVD).
- Additional elements added include a group variable (HeartID), an ID variable (seqnum), Sample Weight, BMI, BMI Category, binaries for female and male, and Obesity Category.
- No PHI/PII revealed in this data set!

ODS Graphics' Place in SAS Graphics Continuum



ODS Graphics' Place in SAS Graphics Continuum

Is the ODS Graphics Statement Necessary?



Q: Is the ODS Graphics statement necessary in producing SAS Graphics?

A. It all depends:

- ODS Graphics is enabled by default for most systems
- Not required for SG procedures
- Is required for Statistical procedures

ODS Graphics' Place in SAS Graphics Continuum

Cue the Sandwich Metaphor

ODS GRAPHICS
<OFF | ON> <
/ options > ;

ods graphics <ON>

/ options;

ods graphics off;

ODS Graphics' Place in SAS Graphics Continuum

Your SAS Environment Matters

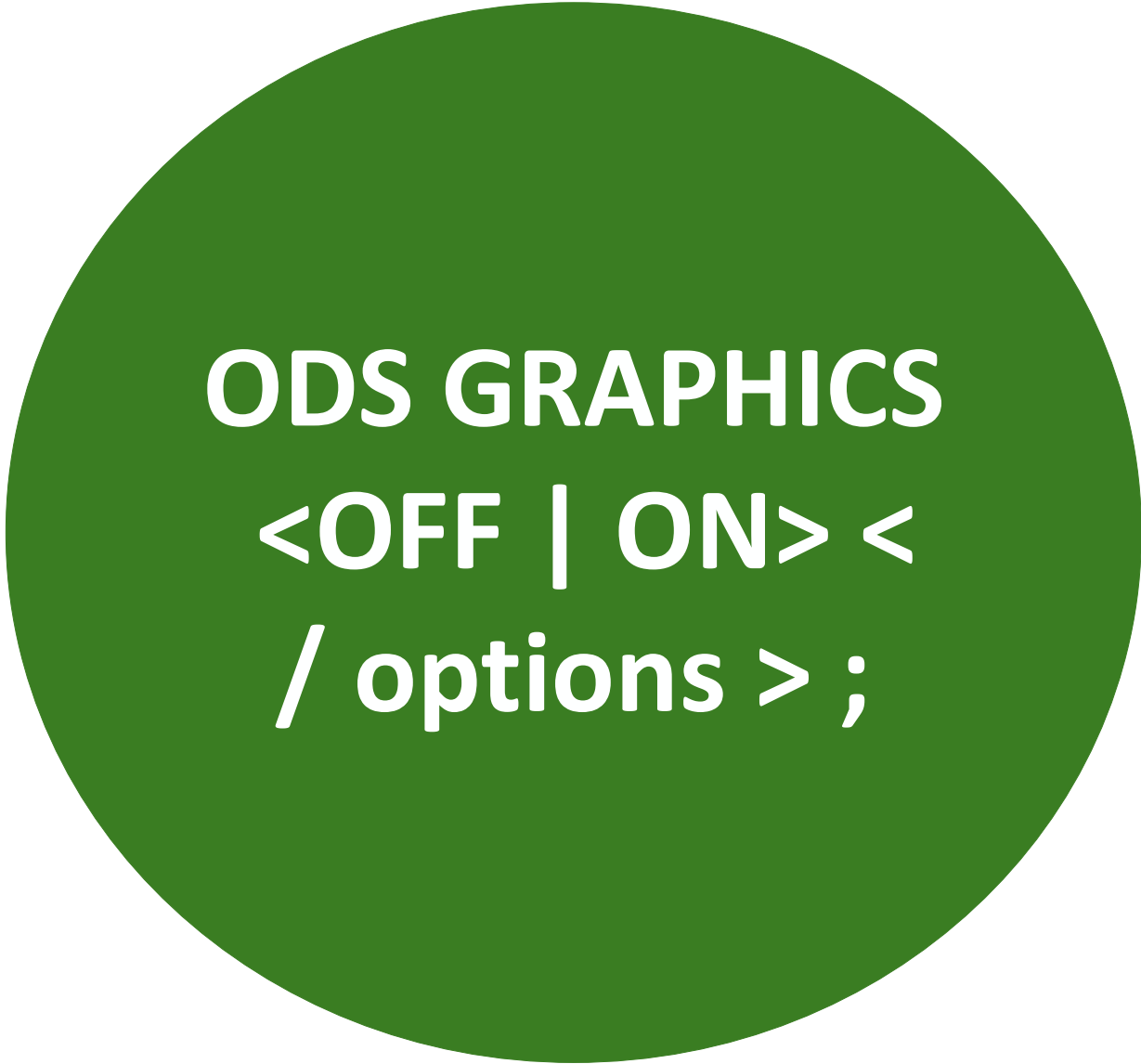
ODS GRAPHICS

**<OFF | ON> <
/ options > ;**

- Where you are running your SAS Code matters to ODS Graphics. For example:
 - In some SAS instances, it is not necessary to use the ODS Graphics statement at all – it is included as part of your start up in some environments running SAS Studio and SAS Enterprise Guide.
 - In all SAS instances, if you do not specify options in the ODS Graphics statement, SAS runs with default values – which may or may not be what you are looking for.
 - Some options may work a little differently in different systems such as the default location for generating graphic files.

ODS Graphics' Place in SAS Graphics Continuum

Your Output Destination Matters

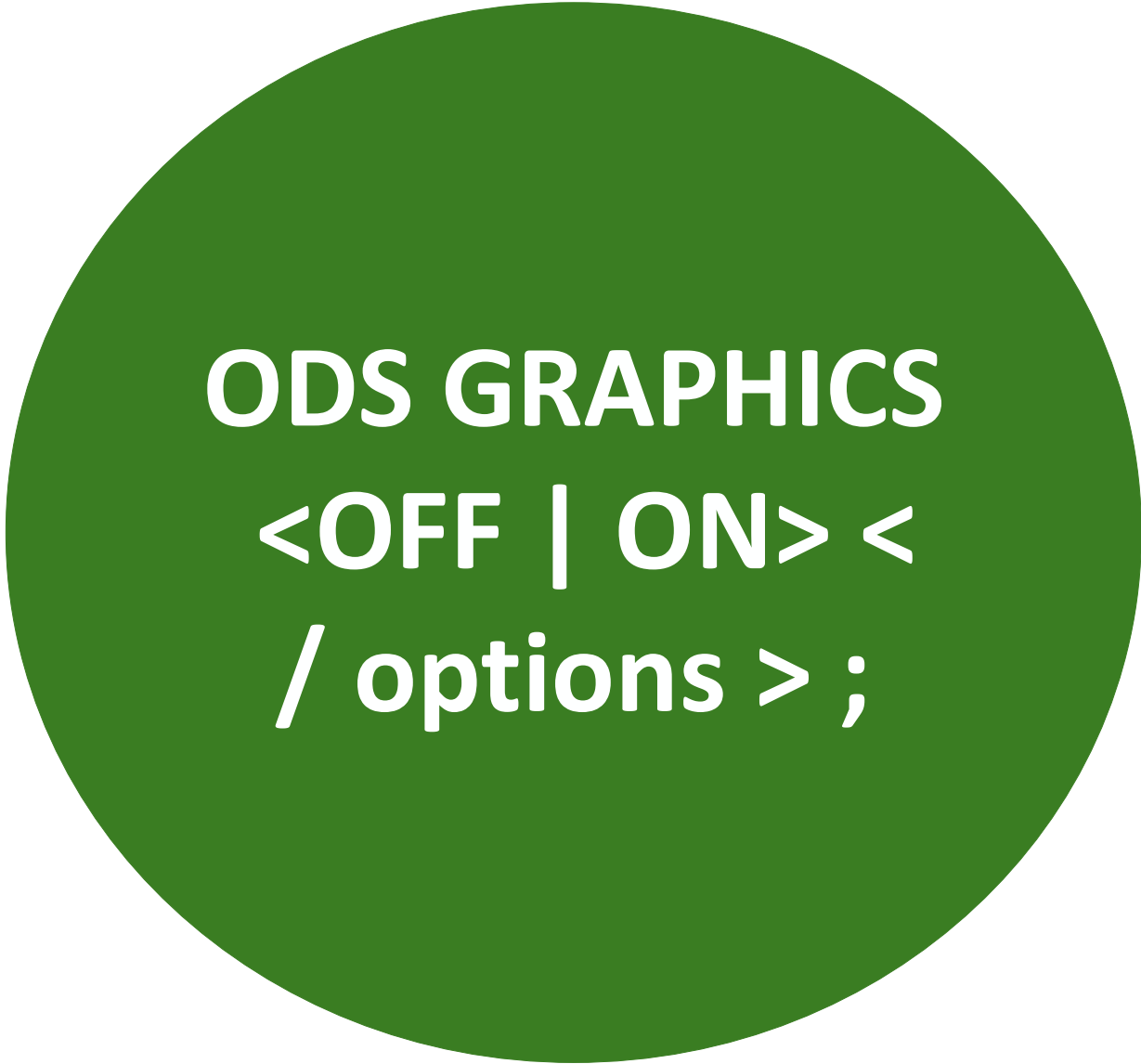


ODS GRAPHICS
<OFF | ON> <
/ options >;

- Your output destination matters to ODS Graphics, even if you are not writing to a typical destination such as RTF, PDF, etc. (but especially matters if you are!)
- Different ODS output destinations have different style templates, different default image types, and different defaults for ODS Graphic Options.. For example:
 - ODS HTML has the default style template of HTMLBlue on my system, default image type of PNG, has the option of adding a “gpath” or “path” to direct created images to,, and has the option of adding tooltips with the `IMAGEMAP=ON` option.
 - ODS RTF has the default style template of RTF on my system, default image type of PNG, and has the option of specifying `image_dpi` (dots per inch).

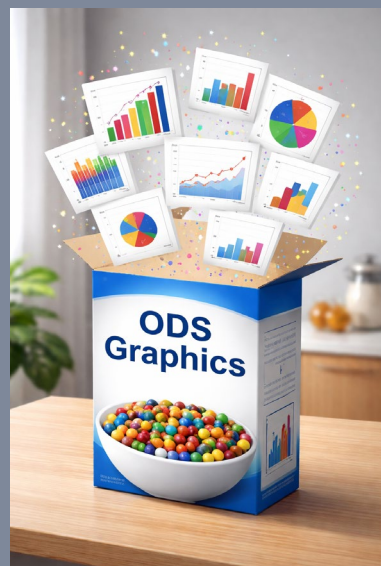
ODS Graphics' Place in SAS Graphics Continuum

Your SAS Procedures and Templates Matter



ODS GRAPHICS
<OFF | ON> <
/ options > ;

- The SAS procedures you are running and SAS templates you are using matters to ODS Graphics. For example:
 - Traditional SAS Statistical Procedures produce different graphic outputs. The SAS documentation on Statistical Procedures will become your best friend. In order to control the output, you need to understand what graphic is produced by each procedure you use.
 - Within each statistical procedure, there often are different options for each graphic type, which override the ODS Graphics statement.
 - If you are running Statistical Graphics (SG) procedures, there are a whole different set of behaviors and options, which also override the ODS Graphics statement.
 - Lastly, any SAS templates (or GTL) have options that will override the ODS Graphics statement.



Simple Is as Simple Does



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Simple Is as Simple Does

How it works

```
ods graphics off;
```

```
ods graphics <on> </ options>;
```

```
proc <procedure> data=<yourdata> <procedural options>;  
    <procedural statement(s)> /  
    <procedural option(s)>  
    plots=(<global-options>)=<plot-request>;  
run;
```

```
ods graphics off;
```

Simple Is as Simple Does

Another layer brings more options into play

```
ods html file="yourdrive\test.html"  
    style=styles.yourstyle;  
ods graphics off;  
ods graphics <on> </ options>;  
proc <procedure> data=<yourdata> <procedural options>;  
    <procedural statement(s)> /  
    <procedural option(s)>  
    plots=(<global-options>)=<plot-request>;  
run;  
ods graphics off;  
ods html close;
```



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Simple Is as Simple Does

Program and Log Snippet

```
ods graphics on;
```

```
]proc surveyfreq data=dash.heart_og order=freq;  
    tables status*(bp_status chol_status sex bmi_cat obesity_cat smoking_status weight_status  
        age_cat height_cat weight_cat chole_cat map_cat) / row chisq plots=all;  
    format bmi_cat bmicatf. obesity_cat obesityf.;  
title2 "Status (Dead or Alive) by Selected SASHELP.HEART variables";  
run;
```

```
ods graphics off;
```

```
145 ods graphics on;  
146  
147 proc surveyfreq data=dash.heart_og order=freq;  
148     tables status*(bp_status chol_status sex bmi_cat obesity_cat smoking_status weight_status  
149         age_cat height_cat weight_cat chole_cat map_cat) / row chisq plots=all;  
150     format bmi_cat bmicatf. obesity_cat obesityf.;  
151 title2 "Status (Dead or Alive) by Selected SASHELP.HEART variables";  
152 run;
```

NOTE: Writing HTML Body file: sashtml1.htm



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Simple Is as Simple Does

“Print” Output

```
ods graphics on;

proc surveyfreq data=dash.heart_og order=freq;
  tables status*(bp_status chol_status sex bmi_cat obesity_cat smoking_status weight_status
    age_cat height_cat weight_cat chole_cat map_cat) / row chisq plots=all;
  format bmi_cat bmicatf. obesity_cat obesityf.;
title2 "Status (Dead or Alive) by Selected SASHELP.HEART variables";
run;

ods graphics off;
```

Table of Status by map_cat						
Status	map_cat	Frequency	Percent	Std Err of Percent	Row Percent	Std Err of Row Percent
Alive	63-92 MAP	1019	19.5623	0.5497	31.6656	0.8201
	111-200 MAP	523	10.0403	0.4164	16.2523	0.6504
	93-100 MAP	899	17.2586	0.5236	27.9366	0.7910
	101-110 MAP	777	14.9165	0.4937	24.1454	0.7545
	Total	3218	61.7777	0.6733	100.0000	
Dead	63-92 MAP	324	6.2200	0.3347	16.2732	0.8273
	111-200 MAP	771	14.8013	0.4921	38.7243	1.0918
	93-100 MAP	392	7.5254	0.3655	19.6886	0.8913
	101-110 MAP	504	9.6756	0.4096	25.3139	0.9746
	Total	1991	38.2223	0.6733	100.0000	
Total	63-92 MAP	1343	25.7823	0.6061		
	111-200 MAP	1294	24.8416	0.5987		
	93-100 MAP	1291	24.7840	0.5983		
	101-110 MAP	1281	24.5921	0.5967		
	Total	5209	100.0000			

Rao-Scott Chi-Square Test	
Pearson Chi-Square	397.5117
Design Correction	1.0000
Rao-Scott Chi-Square	397.5117
DF	3
Pr > ChiSq	<.0001
F Value	132.5039
Num DF	3
Den DF	15624
Pr > F	<.0001
Sample Size = 5209	



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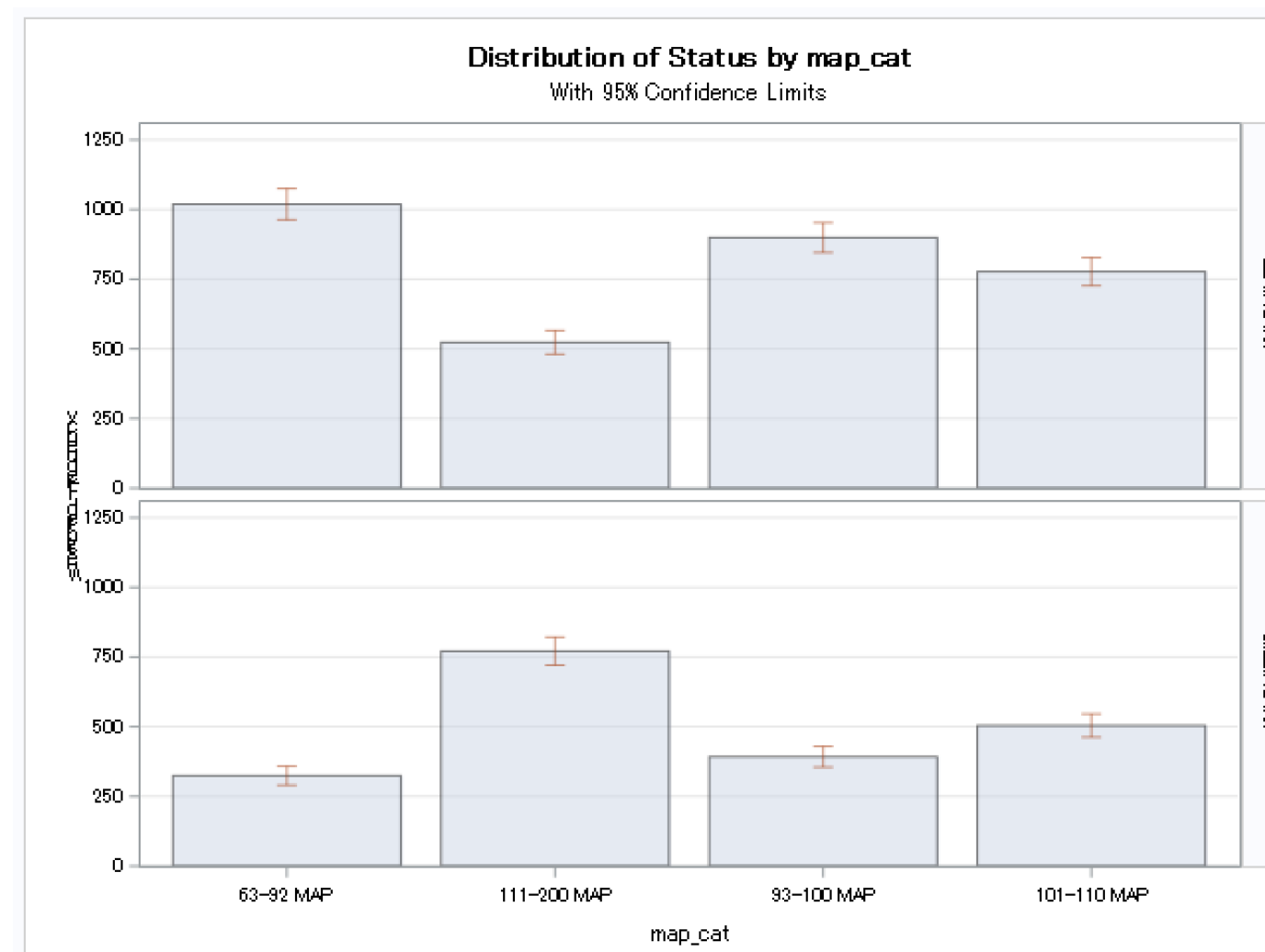
Simple Is as Simple Does

Frequency Plot

```
ods graphics on;
```

```
proc surveyfreq data=dash.heart_og order=freq;  
  tables status*(bp_status chol_status sex bmi_cat obesity_cat smoking_status weight_status  
    age_cat height_cat weight_cat chole_cat map_cat) / row chisq plots=all;  
  format bmi_cat bmicatf. obesity_cat obesityf.;  
title2 "Status (Dead or Alive) by Selected SASHELP.HEART variables";  
run;
```

```
ods graphics off;
```





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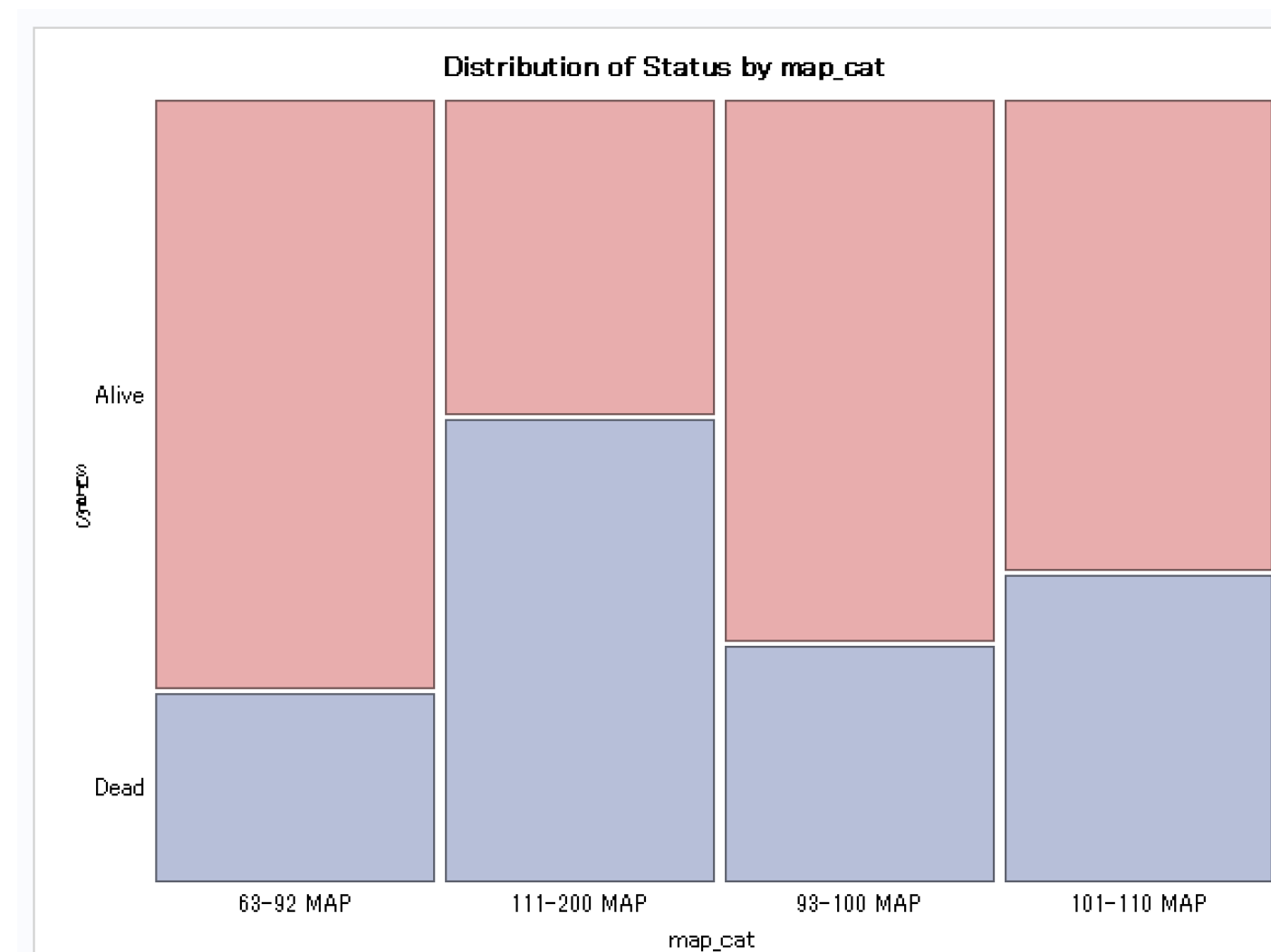
Simple Is as Simple Does

Mosaic Plot

```
ods graphics on;
```

```
proc surveyfreq data=dash.heart_og order=freq;  
  tables status*(bp_status chol_status sex bmi_cat obesity_cat smoking_status weight_status  
    age_cat height_cat weight_cat chole_cat map_cat) / row chisq plots=all;  
  format bmi_cat bmicatf. obesity_cat obesityf.;  
title2 "Status (Dead or Alive) by Selected SASHELP.HEART variables";  
run;
```

```
ods graphics off;
```



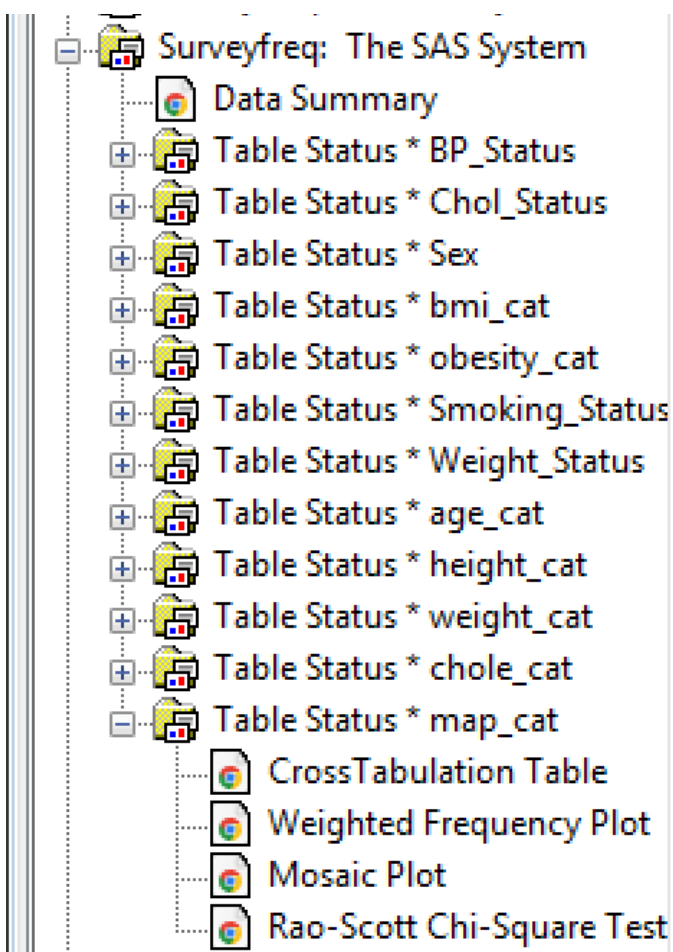
Simple Is as Simple Does

How it looks in the Results Window & Where Are the Graphs?

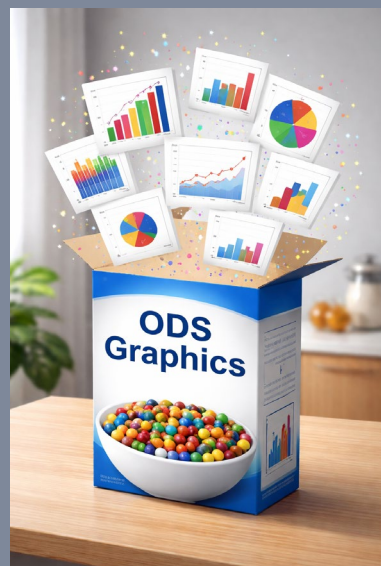
```
ods graphics on;
```

```
proc surveyfreq data=dash.heart_og order=freq;  
  tables status*(bp_status chol_status sex bmi_cat obesity_cat smoking_status weight_status  
    age_cat height_cat weight_cat chole_cat map_cat) / row chisq plots=all;  
  format bmi_cat bmicatf. obesity_cat obesityf.;  
title2 "Status (Dead or Alive) by Selected SASHELP.HEART variables";  
run;
```

```
ods graphics off;
```



Properties of 'MosaicPlot':	
Attribute	Value
Type	HTML
Name	MosaicPlot
Description	Mosaic Plot
Template	Stat.SurveyFreq.Graphics.MosaicPlot
Path	Surveyfreq#2.Table12#1.MosaicPlot#1
Created	4/25/2026 8:45 PM
URL: (or FILE:)	C:\Users\louis\AppData\Local\Temp\SAS Temporary Files_TD23916_GALADRIEL_\sashtml1.htm#IDX47



The ODS Graphics Statement

Options



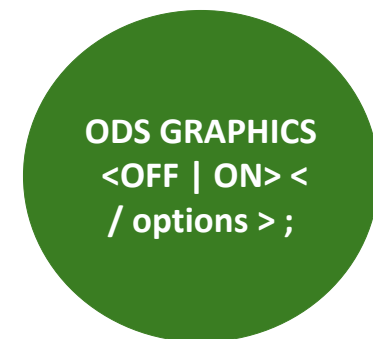
The ODS Graphics Statement – Options



ANTIALIAS	IMAGENAME	OUTPUTFMT
ANTIALIASMAX	INDEX	PANELCELLMAX
ATTRPRIORITY	LABELMAX	SCALE
BORDER	LABELPLACEMENT	SCALEMARKERS
BYLINE	LEGENDAREAMAX	STACKDEPTHMAX
DATASKINMAX	LINEPATTERNOBSMAX	SUBPIXEL
DISCRETEMAX	LOESSOBSMAX	TIPMAX
DRILLTARGET	LOESSOBSMAX	TOTALCELLMAX
GROUPMAX	NBINSMAX	WIDTH
HEIGHT	NXYBINSMAX	
IMAGEMAP	OBSMAX	

The ODS Graphics Statement – Options

Size and Scaling Output



- WIDTH= —set graph width (e.g., 5in)
- HEIGHT= —set graph height
- IMAGEMAP= —enables tooltips (HTML output)
- SCALE= —scale factor for the graph

The default width and height of image files produced with the ODS Graphics statement in place is 640 pixels wide by 480 pixels high (a 4:3 aspect ratio), 6.4 inches by 4.8 inches.

The default imagemap setting is OFF.

The default scale setting is ON— that is, when the size of a graph is changed (e.g., using WIDTH= or HEIGHT=), all graph elements—such as fonts, markers, lines, and titles—are scaled proportionally to fit the new dimensions.



The ODS Graphics Statement – Options

Image Naming and Management



- `IMAGENAME=` —base name of output file
- `IMAGEFMT=` —format (PNG, SVG, JPEG, etc.)
- `RESET=` —resets options (e.g., `reset=all`)

The default imagename is the SAS ODS Graph name (i.e. `Counterplot`, `freqplot`, etc.)

The default imagefmt is PNG (for HTML output), but varies by destination.

SAS automatically appends a number to the imagename if created in the same session to prevent overwriting. To prevent this behavior, you can use `RESET=ALL`, `RESET=INDEX`, etc.

The ODS Graphics Statement – Options

Output Control

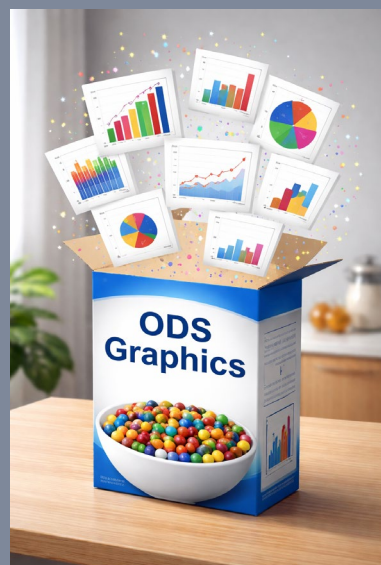


- MAXOBS= —limit observations used in plots
- INDEX= —adds index to image names
- OUTPUTFMT= —controls format in some destinations

Maxobs limits the number of observations used in plots. The default is to use all observations, but some plots will simply not be generated due to too many observations to realistically portray. SAS will suggest a number to use.

Index tells SAS to increment numbers in a giveplotname – which is the default behavior (which can be turned off with RESET=INDEX)

Outputfmt is largely controlled by destinations, but in some instances, you can opt for a different output format. The default outputfmt is dependent on the destination.



SAS Statistical Procedures Producing ODS Graphics

SAS Statistical Procedures

Selected SAS Procedures Supporting ODS Graphics – SAS/STAT



```
proc <procedure>  
plots(<global-options>)  
=<plot-request>;
```

PROC GLM: Plots diagnostic graphs and interaction plots for linear models.

PROC LOGISTIC: Produces ROC curves, odds ratio plots, and predicted probabilities.

PROC LIFETEST: Generates Kaplan-Meier survival curves.

PROC REG: Creates scatter plots, fit plots, and residual diagnostics.

PROC MIXED: Produces diagnostic plots for mixed models.

PROC PHREG: Plots survival and hazard functions.

PROC GENMOD: Generates plots for generalized linear models.

PROC GAM / GAMPL: Plots spline components for generalized additive models.

PROC ROBUSTREG: Produces plots for robust regression analysis.

PROC CORRESP: Generates correspondence analysis plots.

PROC PRINCOMP / PRINQUAL: Plots principal components and results.

PROC KDE: Generates kernel density estimation plots.

PROC LOESS: Plots LOESS curves and diagnostic plots.

PROC MI: Creates plots for missing data imputation.



SAS Statistical Procedures

Selected SAS Procedures Supporting ODS Graphics- SAS/ETS, SAS/QC, SG Procedures

```
proc <procedure>  
plots(<global-options>  
=<plot-request>;
```

SAS/ETS Procedures:

PROC ARIMA: Plots time series data, forecasts, and diagnostic plots.

PROC AUTOREG: Produces autocorrelation and stationarity plots.

PROC VARMAX: Plots vector autoregressive model diagnostics.

PROC TIMESERIES: Creates time series plots.

PROC UCM: Produces unobserved component model plots.

SAS/QC Procedures (Quality Control)

PROC BOXPLOT: Creates high-resolution box plots.

SAS SG Procedures (Statistical Graphics)

PROC SGPLOT: Generates single-celled plots (scatter, box, bar, histograms).

PROC SGSCATTER: Creates multipanel scatter plots.

PROC SGPANEL: Produces paneled plots based on classification variables.

PROC CORR: Generates scatter plots and correlation matrices.

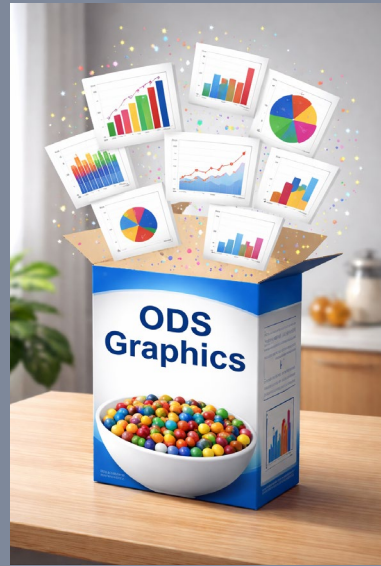
SAS Statistical Procedures

How does the ODS Graphics Statement work with STAT Procedures?



```
proc <procedure>  
plots(<global-options>)  
=<plot-request>;
```

- Each stats procedure produces its own list of plots – some may overlap across procedures, others may not.
- Some stats procedure create graphs without using the ODS Graphics statement, others do not.
- The SAS documentation is invaluable in informing you what plot requests will result in what plots.
- Presented below are some common plots= statements.



SAS Statistical Procedures

Plots= Options



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SAS Statistical Procedures Plots=Options

Plot Requests

- PLOTS=ALL —all available plots
- PLOTS=NONE —suppress plots
- PLOTS=DEFAULT —default set
- PLOTS=RESIDUALPLOT
- PLOTS=DIAGNOSTICS
- PLOTS=EFFECTPLOT
- PLOTS=ROC
- PLOTS=SURVIVAL
- PLOTS=FREQPLOT
- PLOTS=MOSAIC

```
proc <procedure>  
plots(<global-options>  
=<plot-request>;
```

SAS Statistical Procedures Plots=Options

Layout / Orientation

- UNPACK —separate panels into individual plots
- ONLY —restrict to specified plots
- OVERLAY —overlay multiple elements

Paneling and Control

- NROWS= —number of rows in panel
- NCOLS= —number of columns
- BY= —panel by variable (procedure-dependent)

```
proc <procedure>  
plots(<global-options>  
=<plot-request>;
```


SAS Statistical Procedures Plots Requests

Plot Specific Suboptions

```
plots (only) = effectplot (sliceby= gender plotby= region) ;
```



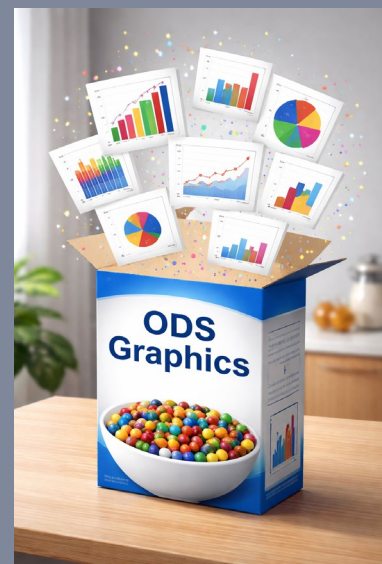
```
proc <procedure>  
plots(<global-options>  
=<plot-request>;
```

Axis / Orientation-like Controls

- ORIENT=HORIZONTAL
- ORIENT=VERTICAL
- FLIP (in some procedures)

Display Options

- CL —confidence limits
- NOBS —show counts
- SMOOTH
- LABEL



Putting the ODS Graphics Statement Through its Paces



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Putting on the Ritz

Without ODS Graphics Statement

```
*****;  
*** PROC FREQ without ODS GRAPHICS ***;  
*****;  
  
ods graphics off; /*this is to make sure no prior statement has turned ods graphics on */  
  
ods trace on;  
proc surveyfreq data=dash.heart_og order=freq;  
    tables status*(bp_status chol_status sex bmi_cat obesity_cat smoking_status weight_status  
        age_cat height_cat weight_cat chole_cat map_cat) / row chisq;  
    format bmi_cat bmicatf. obesity_cat obesityf.;  
title2 "Status (Dead or Alive) by Selected SASHELP.HEART variables";  
run;  
ods trace off;  
  
ods html close;
```

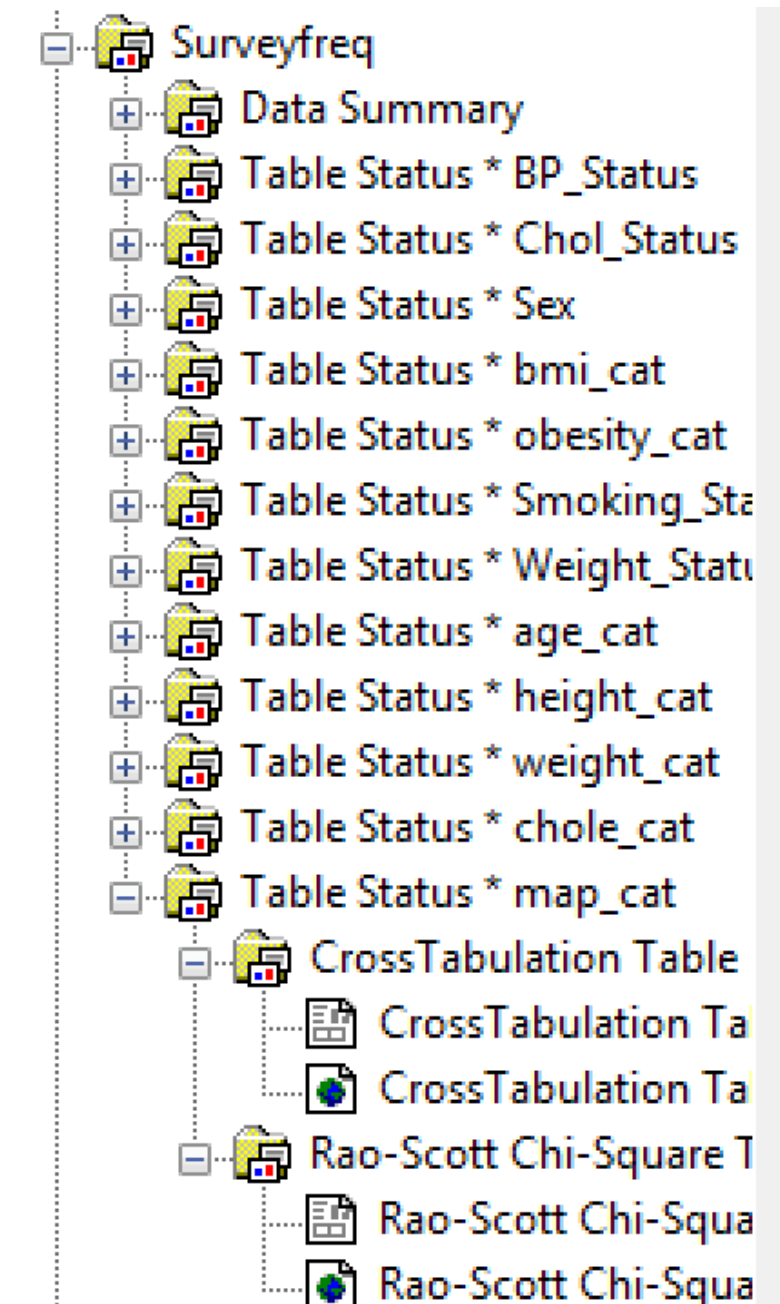
Properties of 'CrossTabs':	
Attribute	Value
Type	Listing
Name	CrossTabs
Description	CrossTabulation Table
Template	Stat.SurveyFreq.CrossTabFreqs
Path	Surveyfreq#1.Table12#1.CrossTabs#1
Created	4/26/2026 7:06 PM
SAS Output: Begins at line	2650
SAS Output: Ends at line	2680

Surveyfreq	
Data Summary	
+ Table Status * BP_Status	
+ Table Status * Chol_Status	
+ Table Status * Sex	
+ Table Status * bmi_cat	
+ Table Status * obesity_cat	
+ Table Status * Smoking_Status	
+ Table Status * Weight_Status	
+ Table Status * age_cat	
+ Table Status * height_cat	
+ Table Status * weight_cat	
+ Table Status * chole_cat	
+ Table Status * map_cat	
CrossTabulation Table	
Rao-Scott Chi-Square Test	

Putting on the Ritz

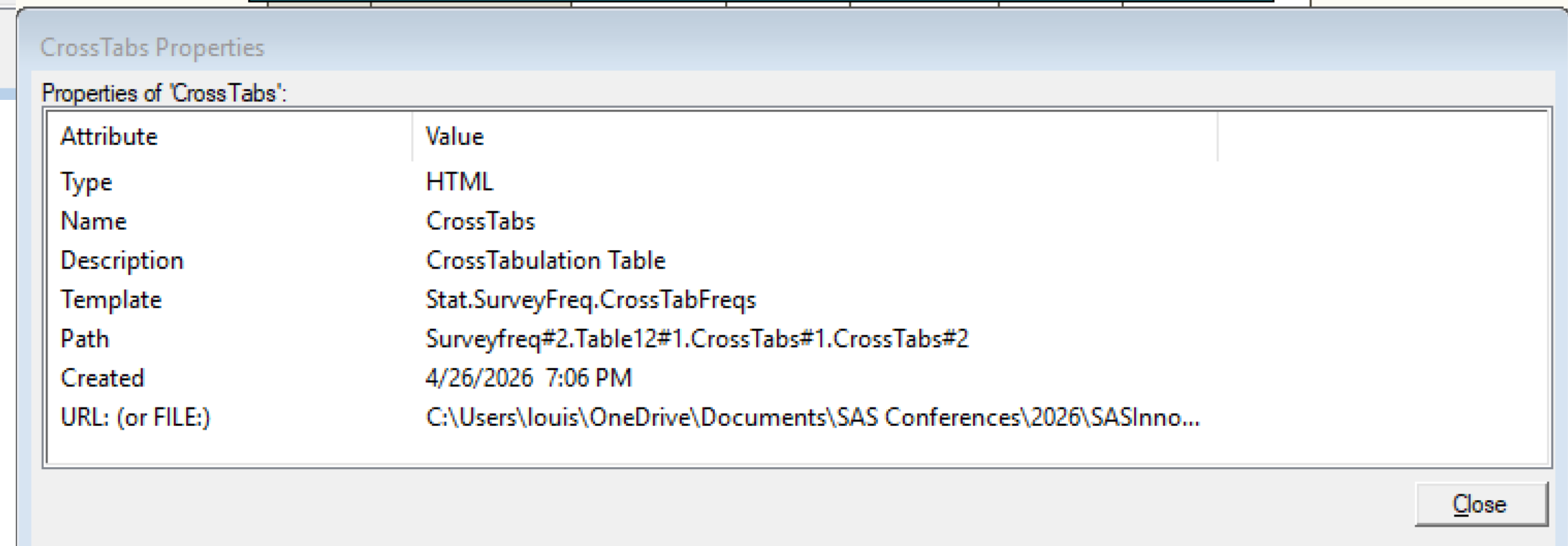
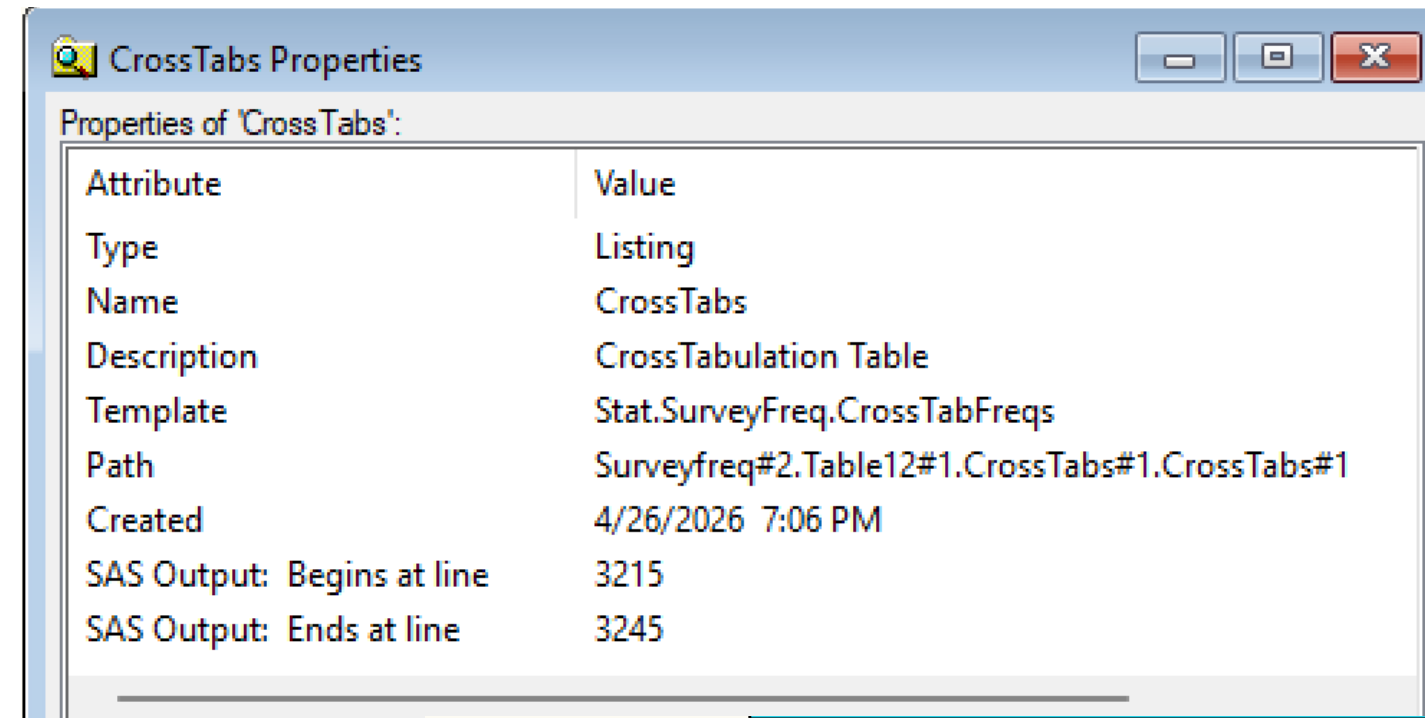
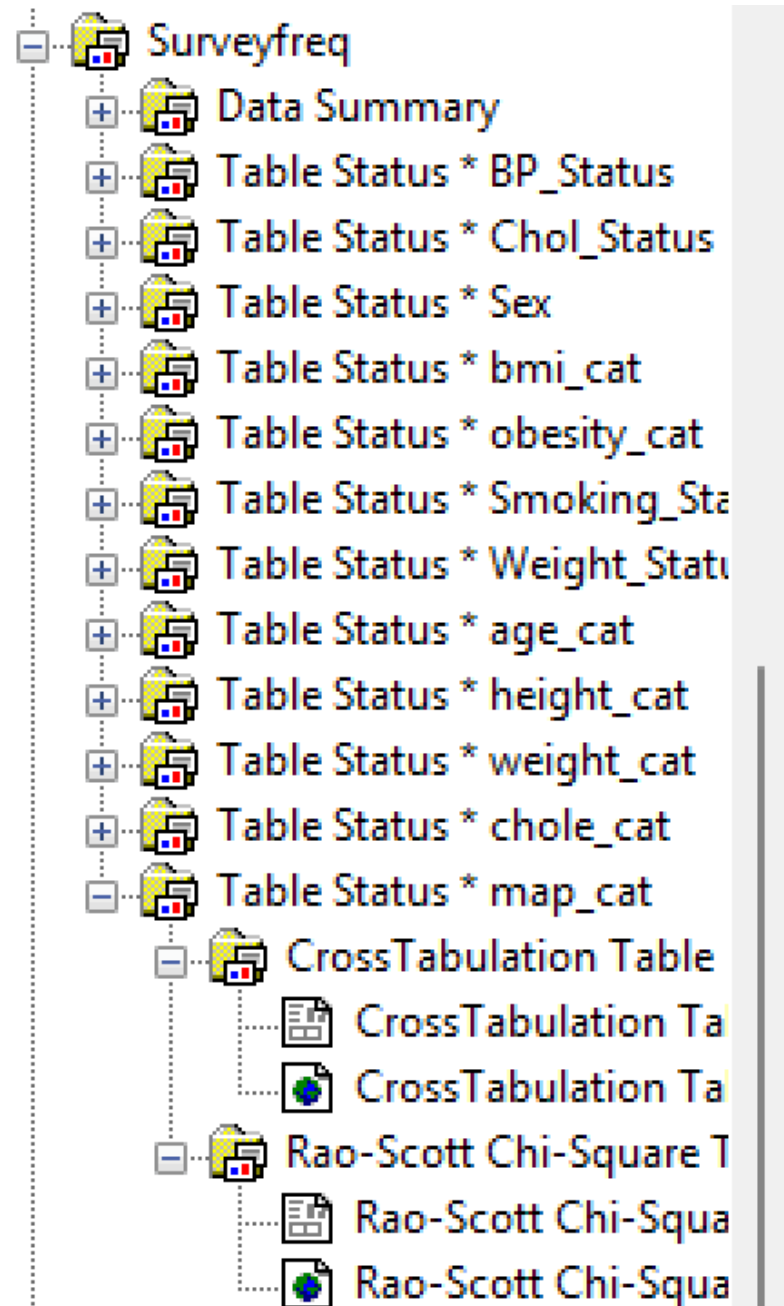
With ODS HTML, With ODS Graphics Statement, Without Plot Statement

```
ods html file="C:\Users\louis\OneDrive\Documents\SAS Conferences\2026\SASInnovate\ODSGraphicsState;  
*****;  
*** PROC FREQ with ODS GRAPHICS ON / no plot statement ***;  
*****;  
  
ods graphics on;  
  
ods graphics / reset=index;  
ods graphics / reset=all;  
  
ods trace on;  
  
proc surveyfreq data=dash.heart_og order=freq;  
    tables status*(bp_status chol_status sex bmi_cat obesity_cat smoking_status weight_status  
        age_cat height_cat weight_cat chole_cat map_cat) / row chisq;  
    format bmi_cat bmicatf. obesity_cat obesityf.;  
    title2 "Status (Dead or Alive) by Selected SASHELP.HEART variables";  
run;  
  
ods graphics off;  
  
ods trace off;  
  
ods html close;
```



Putting on the Ritz

With ODS HTML, With ODS Graphics Statement, Without Plot Statement



Putting on the Ritz

With ODS HTML, With ODS Graphics Statement, Without Plot Statement

Surveyfreq
Data Summary
Table Status * BP_Status
Table Status * Chol_Status
Table Status * Sex
Table Status * bmi_cat
Table Status * obesity_cat
Table Status * Smoking_Sta
Table Status * Weight_Statu
Table Status * age_cat
Table Status * height_cat
Table Status * weight_cat
Table Status * chole_cat
Table Status * map_cat
CrossTabulation Table
CrossTabulation Ta
CrossTabulation Ta
Rao-Scott Chi-Square T
Rao-Scott Chi-Squa
Rao-Scott Chi-Squa

Status (Dead or Alive) by Selected SASHELP.HEART variables

Table of Status by map_cat

Status	map_cat	Frequency	Percent	Std Err of Percent	Row Percent	Std Err of Row Percent
Alive	63-92 MAP	1019	19.5623	0.5497	31.6656	0.8201
	111-200 MAP	523	10.0403	0.4164	16.2523	0.6504
	93-100 MAP	899	17.2586	0.5236	27.9366	0.7910
	101-110 MAP	777	14.9165			
	Total	3218	61.7777			
Dead	63-92 MAP	324	6.2200			
	111-200 MAP	771	14.8013			
	93-100 MAP	392	7.5254			
	101-110 MAP	504	9.6756			
	Total	1991	38.2223			
Total	63-92 MAP	1343	25.7823			
	111-200 MAP	1294	24.8416			
	93-100 MAP	1291	24.7840			
	101-110 MAP	1281	24.5921			
	Total	5209	100.0000			

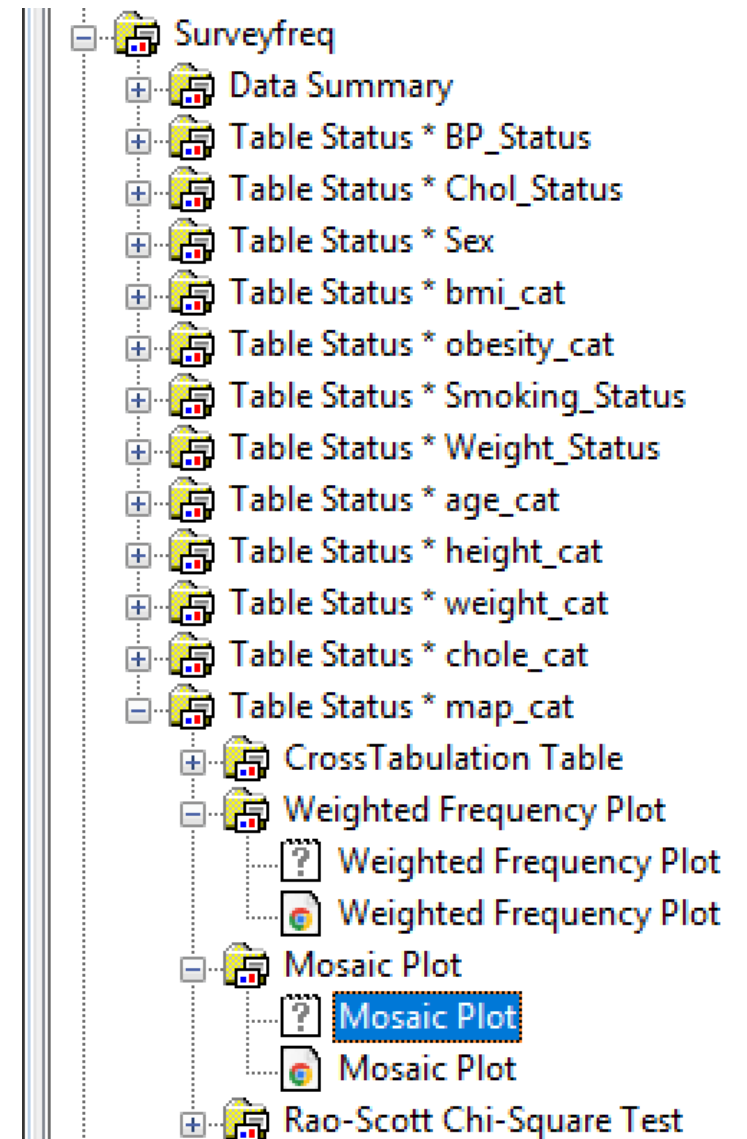
Table of Status by map_cat

Status	map_cat	Frequency	Percent	Std Err of Percent	Row Percent	Std Err of Row Percent
Alive	63-92 MAP	1019	19.5623	0.5497	31.6656	0.8201
	111-200 MAP	523	10.0403	0.4164	16.2523	0.6504
	93-100 MAP	899	17.2586	0.5236	27.9366	0.7910
	101-110 MAP	777	14.9165	0.4937	24.1454	0.7545
	Total	3218	61.7777	0.6733	100.0000	
Dead	63-92 MAP	324	6.2200	0.3347	16.2732	0.8273
	111-200 MAP	771	14.8013	0.4921	38.7243	1.0918
	93-100 MAP	392	7.5254	0.3655	19.6886	0.8913
	101-110 MAP	504	9.6756	0.4096	25.3139	0.9746
	Total	1991	38.2223	0.6733	100.0000	
Total	63-92 MAP	1343	25.7823	0.6061		
	111-200 MAP	1294	24.8416	0.5987		
	93-100 MAP	1291	24.7840	0.5983		
	101-110 MAP	1281	24.5921	0.5967		
	Total	5209	100.0000			

Putting on the Ritz

With ODS HTML, With ODS Graphics Statement, With Plot Statement

```
*** PROC FREQ with ODS GRAPHICS ON / plot statement ***;  
*****;  
  
ods graphics on;  
  
ods graphics / reset=index;  
ods graphics / reset=all;  
  
ods trace on;  
  
ods graphics off;  
  
ods html file="PROCFREQwODSGraphicsWGPATH.html" gpath="&odsdir" style=styles.analysis;  
ods graphics on;  
  
proc surveyfreq data=dash.heart_og order=freq;  
    tables status*(bp_status chol_status sex bmi_cat obesity_cat smoking_status weight_status  
        age_cat height_cat weight_cat chole_cat map_cat) / row chisq plots=all;  
    format bmi_cat bmicatf. obesity_cat obesityf.;  
    title2 "Status (Dead or Alive) by Selected SASHELP.HEART variables";  
run;  
  
ods graphics off;  
ods html close;  
  
ods trace off;  
  
ods html close;
```



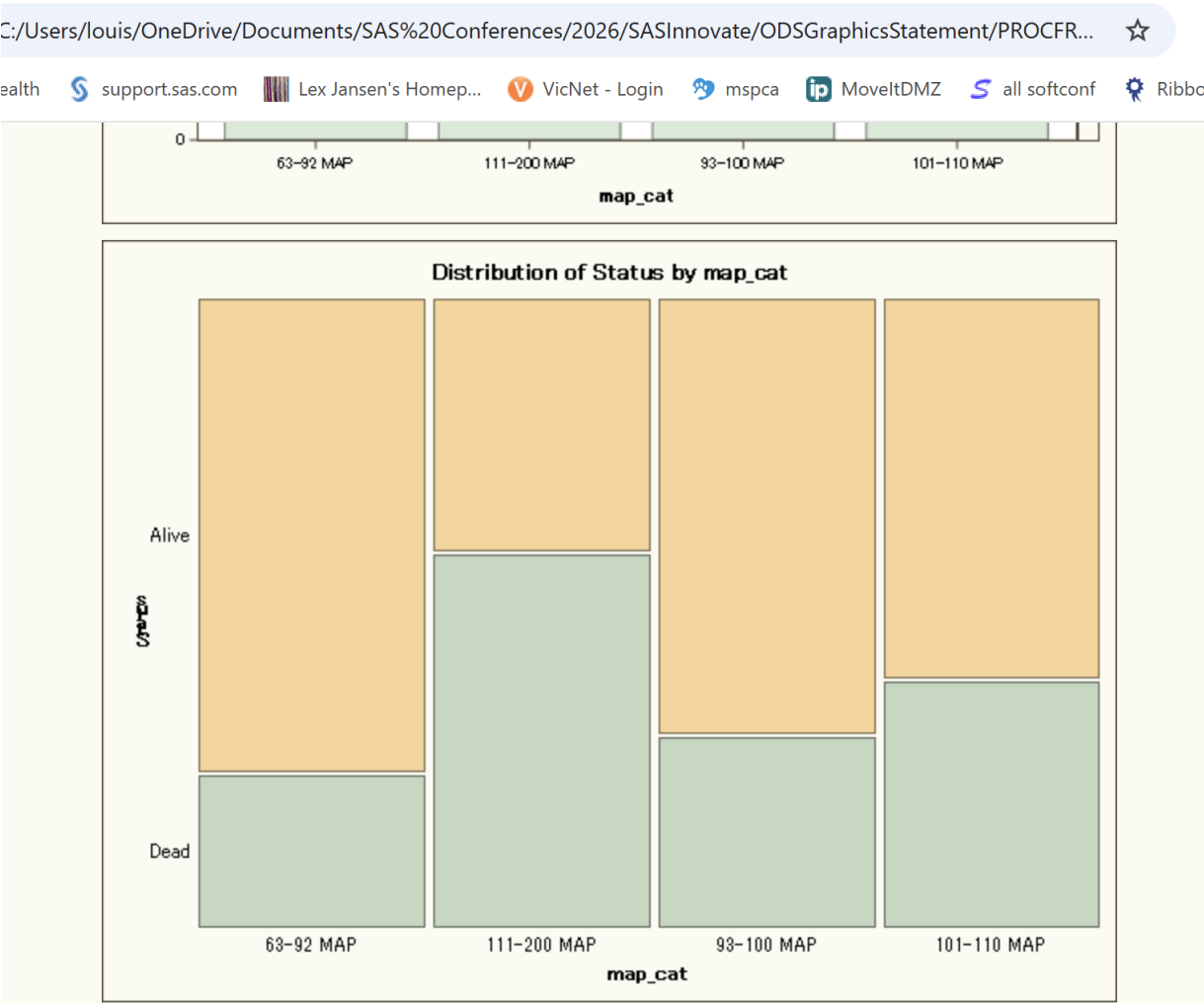
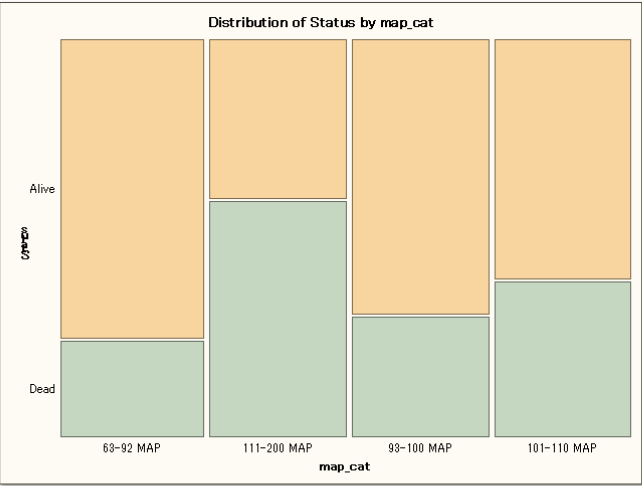


480px
480px

Putting on the Ritz

With ODS HTML, With ODS Graphics Statement, With Plot Statement

MosaicPlot1.png	✓	4/26/2026 4:48 PM	PNG File
MosaicPlot3.png	✓	4/26/2026 4:48 PM	PNG File
MosaicPlot5.png	✓	4/26/2026 4:48 PM	PNG File
MosaicPlot7.png	✓	4/26/2026 4:48 PM	PNG File
MosaicPlot9.png	✓	4/26/2026 4:48 PM	PNG File
MosaicPlot11.png	✓	4/26/2026 4:48 PM	PNG File
MosaicPlot13.png	✓	4/26/2026 4:48 PM	PNG File
MosaicPlot15.png	✓	4/26/2026 4:48 PM	PNG File
MosaicPlot17.png	✓	4/26/2026 4:48 PM	PNG File
MosaicPlot19.png	✓	4/26/2026 4:48 PM	PNG File
MosaicPlot21.png	✓	4/26/2026 4:48 PM	PNG File
MosaicPlot23.png	✓	4/26/2026 4:48 PM	PNG File



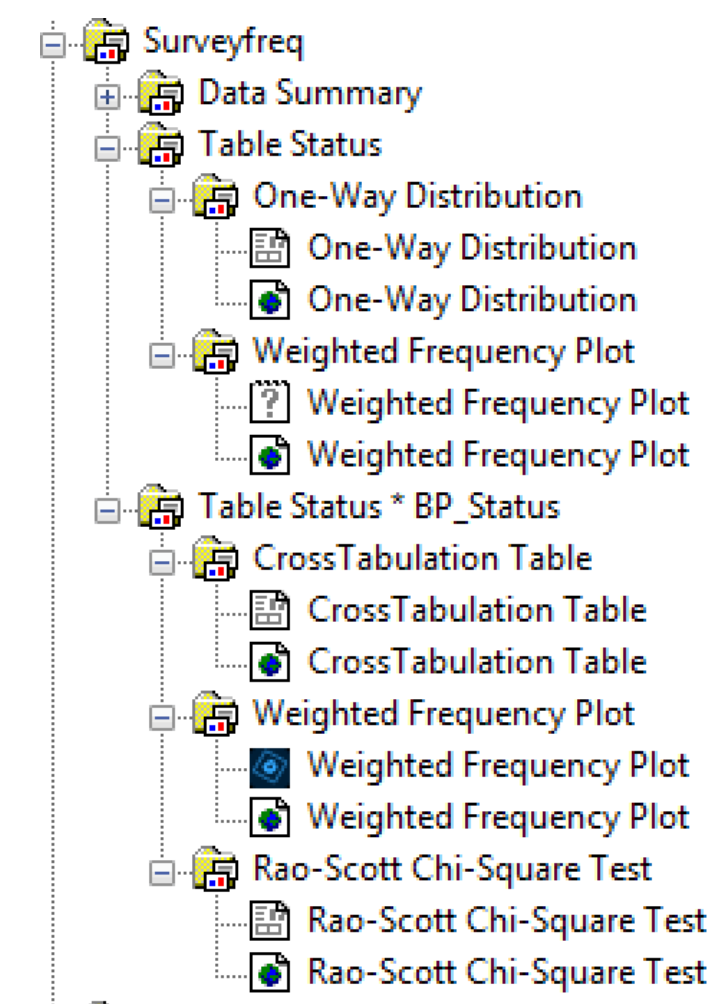


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Putting on the Ritz

Adding in some options

```
ods html file="PROCFREQwODSGraphicsWGPATHnOPTIONSONeVar.html" gpath="&odsdir" style=styles.analysis;  
*****;  
*** PROC FREQ with ODS GRAPHICS ON / plot and options ***;  
*****;  
  
ods graphics on;  
ods graphics / reset=index;  
ods graphics / reset=all imagename="StatusVSBPStatus" border=off imagemap=on;  
ods trace on;  
  
proc surveyfreq data=dash.heart_og order=freq;  
    tables status / plots=freqplot;  
    tables status*bp_status / row chisq plots=freqplot;  
    format bmi_cat bmicatf. obesity_cat obesityf.;  
title2 "Status (Dead or Alive) by Selected SASHELP.HEART variables";  
run;  
  
ods graphics off;  
ods trace off;  
ods html close;  
quit;
```





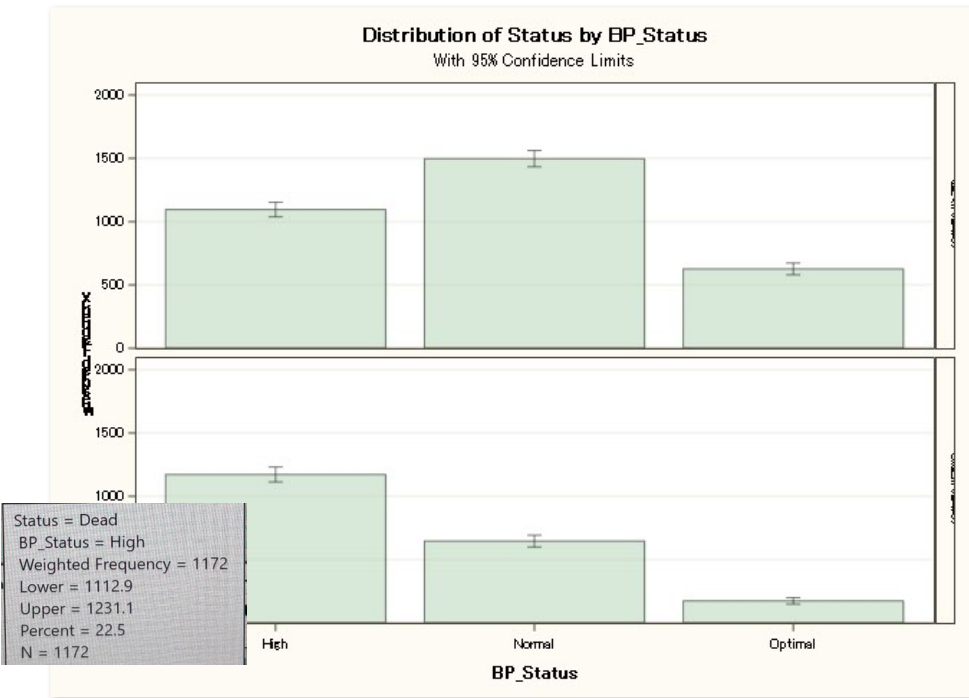
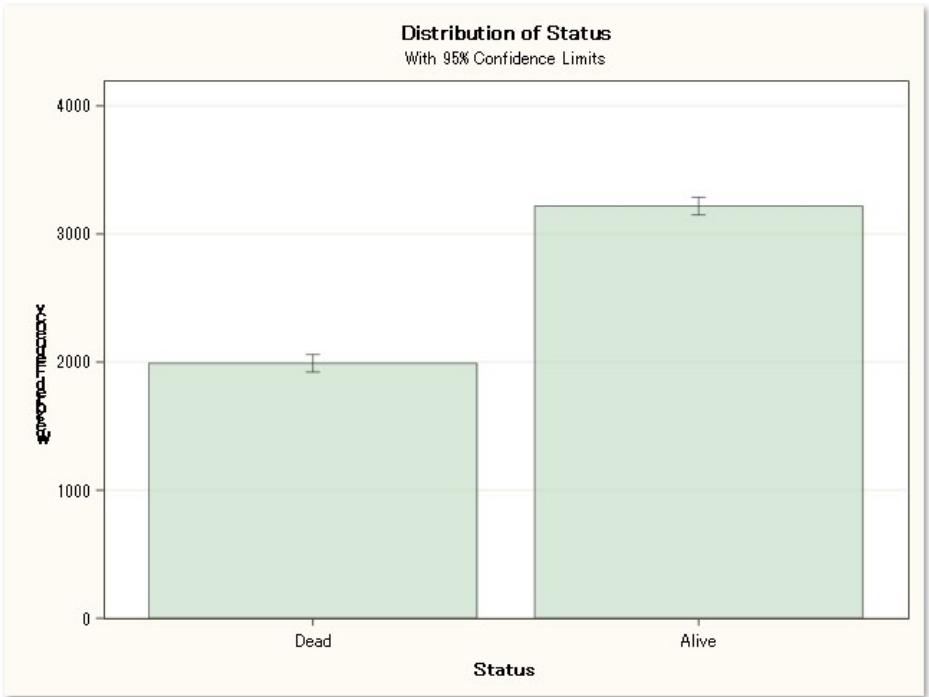
480px
480px

Putting on the Ritz

Adding in some options

StatusVSBPStatus1.png	✓	4/26/20
StatusVSBPStatus3.png	✓	4/26/20
WtFreqPlot1.png	✓	4/26/20
WtFreqPlot3.png	✓	4/26/20
WtFreqPlot5.png	✓	4/26/20
WtFreqPlot7.png	✓	4/26/20
WtFreqPlot9.png	✓	4/26/20
WtFreqPlot11.png	✓	4/26/20

StatusVSBPStatus3.png	✓	4/26/20
WtFreqPlot1.png	✓	4/26/20
WtFreqPlot3.png	✓	4/26/20
WtFreqPlot5.png	✓	4/26/20
WtFreqPlot7.png	✓	4/26/20
WtFreqPlot9.png	✓	4/26/20
WtFreqPlot11.png	✓	4/26/20
WtFreqPlot13.png	✓	4/26/20



Putting on the Ritz

Macro Routine to Create Individual Named Graphics

```
%macro odsgraph(iname=StatusVSBPStatus,crossvar=BPStatus,tit2=BP Status);

ods graphics on;
ods graphics / reset=index;
ods graphics / reset=all
    imagename="C:\Users\louis\OneDrive\Documents\SAS Conferences\2026\SASInnovate\ODSGraphicsStatement\&iname."
    border=off imagemap=on;
ods trace on;

proc surveyfreq data=dash.heart_og order=freq;
    tables status*&crossvar / row chisq plots=freqplot;
    format bmi_cat bmicatf. obesity_cat obesityf.;
title2 "Status (Dead or Alive) by &tit2.";
run;

ods graphics off;
ods trace off;

%mend;
```

Putting on the Ritz

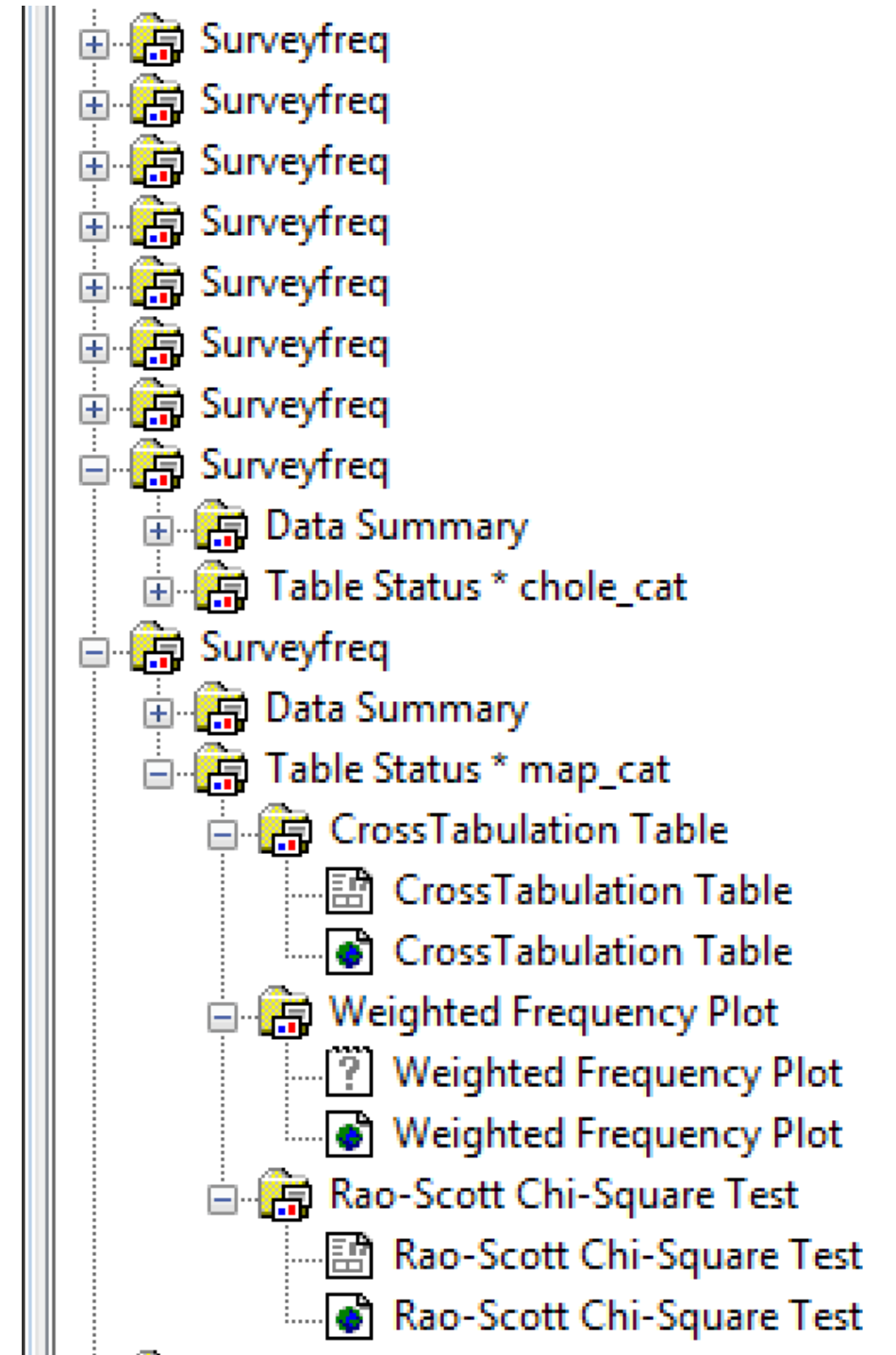
Macro Routine to Create Individual Named Graphics

```
ods html file="PROCFREQwODSGraphicsWGPATHnOPTIONSAllVars.html" gpath="&odsdir" style=styles.analysis;

*****;
*** PROC FREQ with ODS GRAPHICS ON / plot and options macro ***;
*****;
%odsgraph(iname=StatusVSBPStatus,crossvar=BPStatus,tit2=BP Status);
%odsgraph(iname=StatusVScholStatus,crossvar=chol_Status,tit2=Cholesterol Status);
%odsgraph(iname=StatusVSSex,crossvar=Sex,tit2=Sex);
%odsgraph(iname=StatusVsbmi_cat,crossvar=bmi_cat,tit2=BMI Category);
%odsgraph(iname=StatusVSObesityCategory,crossvar=obesity_cat,tit2=Obesity Category);
%odsgraph(iname=StatusVSsmoking,crossvar=smoking_status,tit2=Smoking);
%odsgraph(iname=StatusVSWeightStatus,crossvar=weight_status,tit2=Weight Status);

%odsgraph(iname=StatusVSAgeCat,crossvar=age_cat,tit2=Age Quartile);
%odsgraph(iname=StatusVSHtCat,crossvar=height_cat,tit2=Height Quartile);
%odsgraph(iname=StatusVSWgtCat,crossvar=weight_cat,tit2=Weight Quartile);
%odsgraph(iname=StatusVSCholeCat,crossvar=chole_cat,tit2=Cholesterol Quartile);
%odsgraph(iname=StatusVSMAPCat,crossvar=map_cat,tit2=MAP Category);

ods html close;
```



Putting on the Ritz

Using Plots suboptions – bar, orientation, scale

```
ods trace on;  
ods graphics / reset=index;  
ods graphics / reset=all imagename="SmokingStatus1" border=off imagemap=on;
```

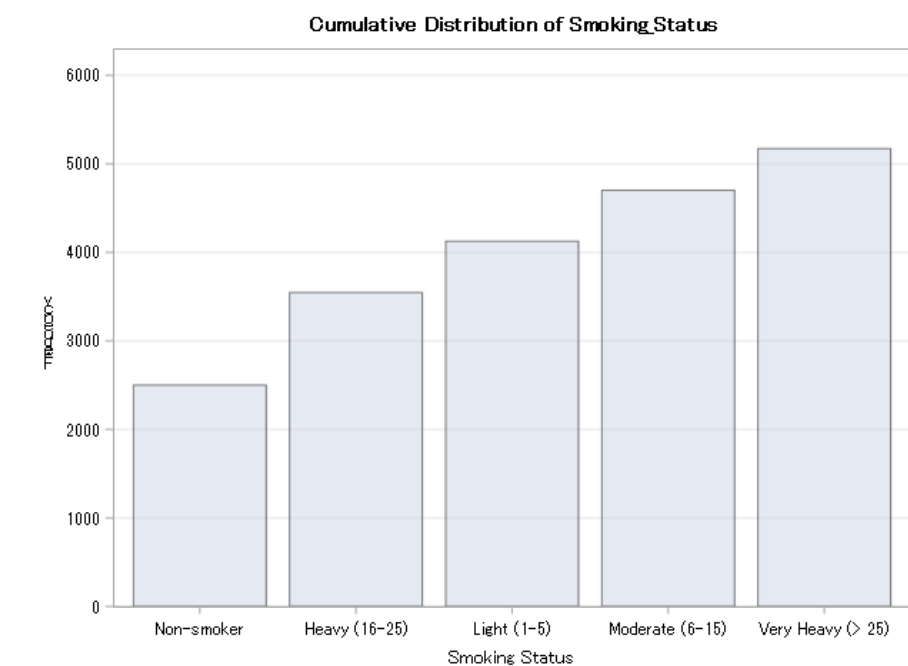
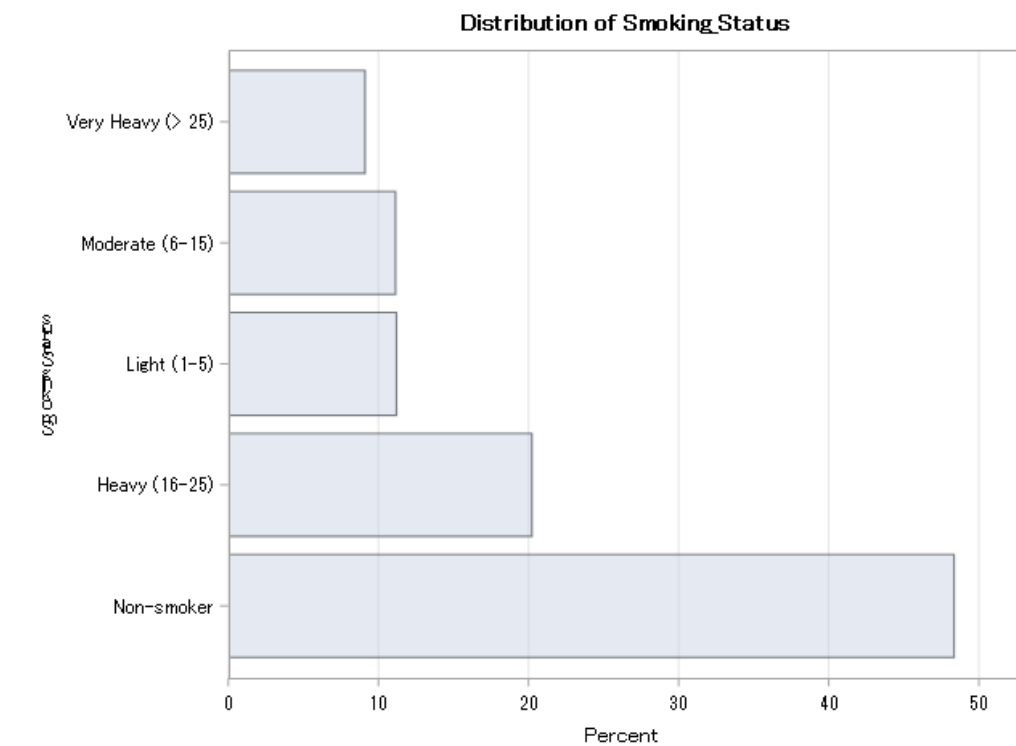
```
proc freq data=dash.heart_og order=freq;  
  tables smoking_status / plots=freqplot (type=bar scale=percent orient=horizontal );  
  title2 "Smoking Status - bar, percent, horizontal";  
run;
```

```
ods graphics off;
```

```
ods trace on;  
ods graphics / reset=index;  
ods graphics / reset=all imagename="SmokingStatus2" border=off imagemap=on;
```

```
proc freq data=dash.heart_og order=freq;  
  tables smoking_status / plots=cumfreqplot (type=bar scale=freq orient=vertical );  
  title2 "Smoking Status - bar, count, vertical";  
run;
```

```
ods graphics off;  
ods trace off;
```



Putting on the Ritz

Using Plots suboptions – dot plot, orientation, scale, cumfreqplot

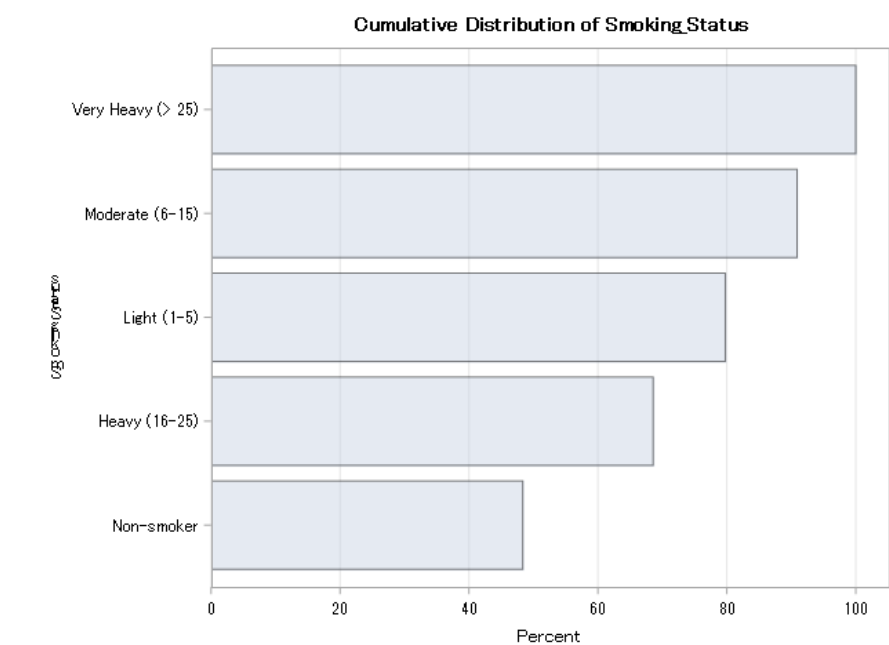
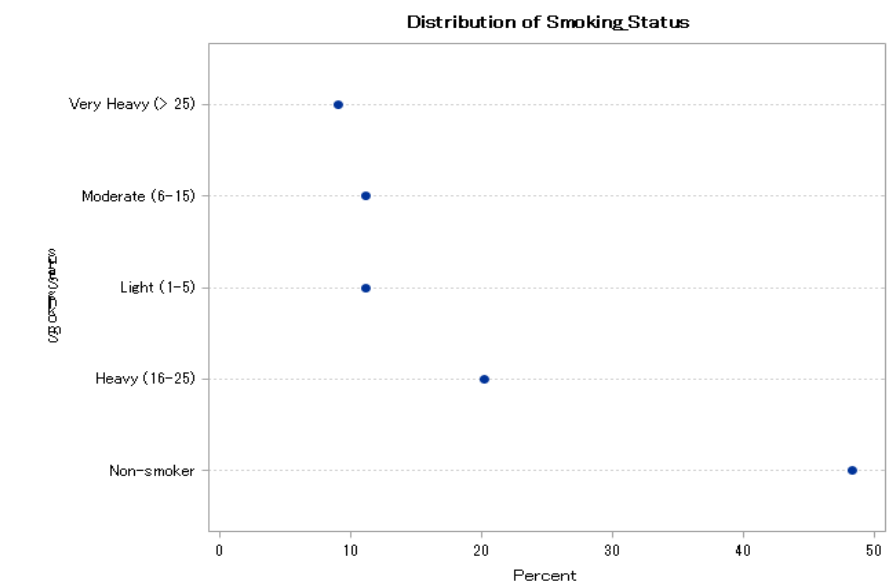
```
ods trace on;
ods graphics / reset=index;
ods graphics / reset=all imagename="SmokingStatus3" border=off imagemap=on;
```

```
proc freq data=dash.heart_og order=freq;
  tables smoking_status / plots=freqplot (type=dot scale=percent orient=horizontal );
  title2 "Smoking Status";
run;
```

```
ods graphics off;
ods trace off;
```

```
ods graphics on;
ods graphics / reset=index;
ods graphics / reset=all imagename="SmokingStatus4" border=off imagemap=on;
```

```
proc freq data=dash.heart_og order=freq;
  tables smoking_status / plots=cumfreqplot (type=bar scale=percent orient=horizontal );
  title2 "Smoking Status ";
run;
```

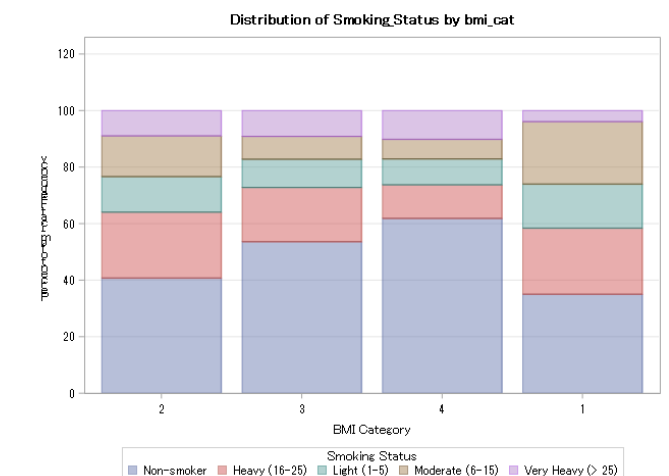


Putting on the Ritz

Using Plots suboptions – Stacked Bar, Group Percent, Mosaic Plot, Colorstat

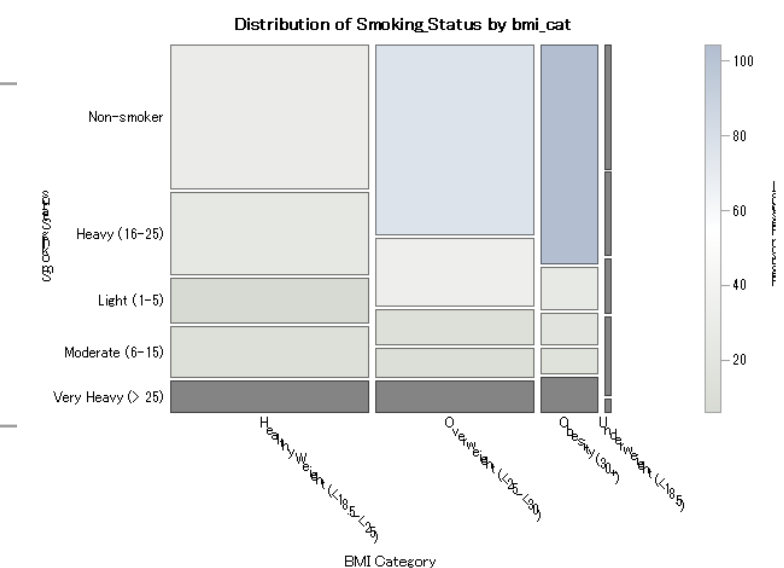
```
ods graphics / reset=index;
ods graphics / reset=all imagename="SmokingStatusBMICAT5" border=off ;
```

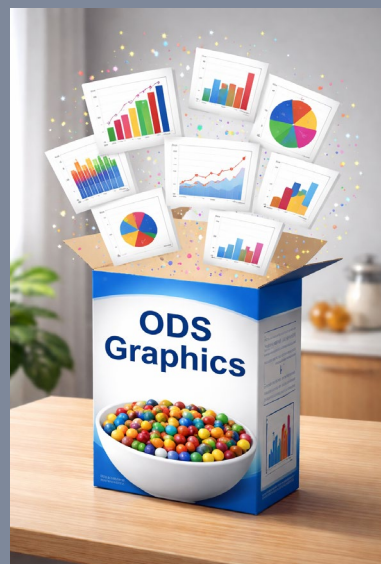
```
proc freq data=dash.heart_og order=freq ;
  tables smoking_status*bmi_cat / plots=freqplot (groupby=column orient=vertical twoway=stacked scale=grouppercent
  title2 "Smoking Status by BMI Category";
run;
```



```
ods graphics / reset=index;
ods graphics / reset=all imagename="SmokingStatusBMICAT6" border=off ;
```

```
proc freq data=dash.heart_og order=freq ;
  tables smoking_status*bmi_cat / plots=mosaicplot (colorstat=pearsonres );
  format bmi_cat bmi_catf.;
  title2 "Smoking Status by BMI Category";
run;
```





Conclusion

Conclusion

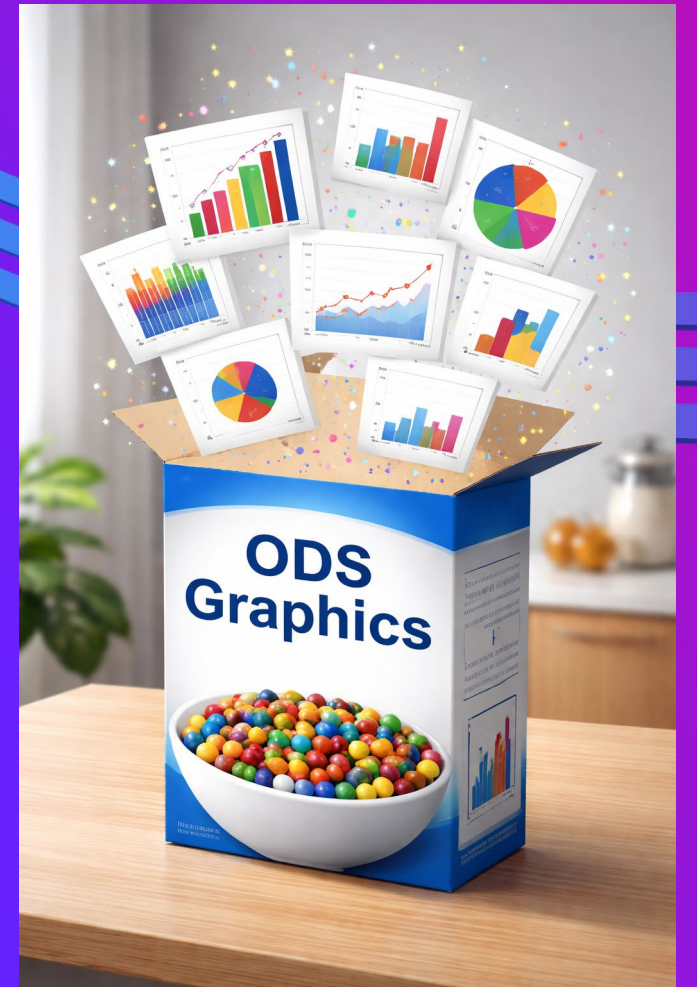
In conclusion, I hope you've enjoyed the fantastic voyage of seeing how much you can do with the ODS Graphics statement and Statistical Procedures, without writing SAS/Graph code, SG procedure code, or GTL code. There is definitely an opportunity for quick and dirty exploratory analyses using graphics output.

What a great time to think “in the box”!



Contact me:
saslouisehadden@gmail.com
Paper, Sample Program, and
Presentation available

<https://github.com/TheGirlWiththeSASTattoo>



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2026